DBMS PROJECT · CSE DEPARTMENT

Knowledge Hub System

Aweb-basedonline bookstore poweredbyastructuredrelationaldatabase — designed for efficient data management, real-time operations, and seamless user experience.

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01 — PROBLEM STATEMENT

The Data Management Challenge

Core Issues

- Large volumes of unstructured data — books, users, orders, payments
- Data redundancy and inconsistency across records
- Slow, inefficient queries without proper indexing
- No transactional integrity for critical operations

The Need

A structured relational database system to centralize data, enforce integrity, and support real-time operations for an online bookstore platform.

02 — OBJECTIVES & OVERVIEW

What We Set Out to Build

Efficient Storage

Optimized schema for fast, reliable data retrieval across all entities

Data Integrity

Referential integrity via foreign keys and constraints; normalized to 3NF

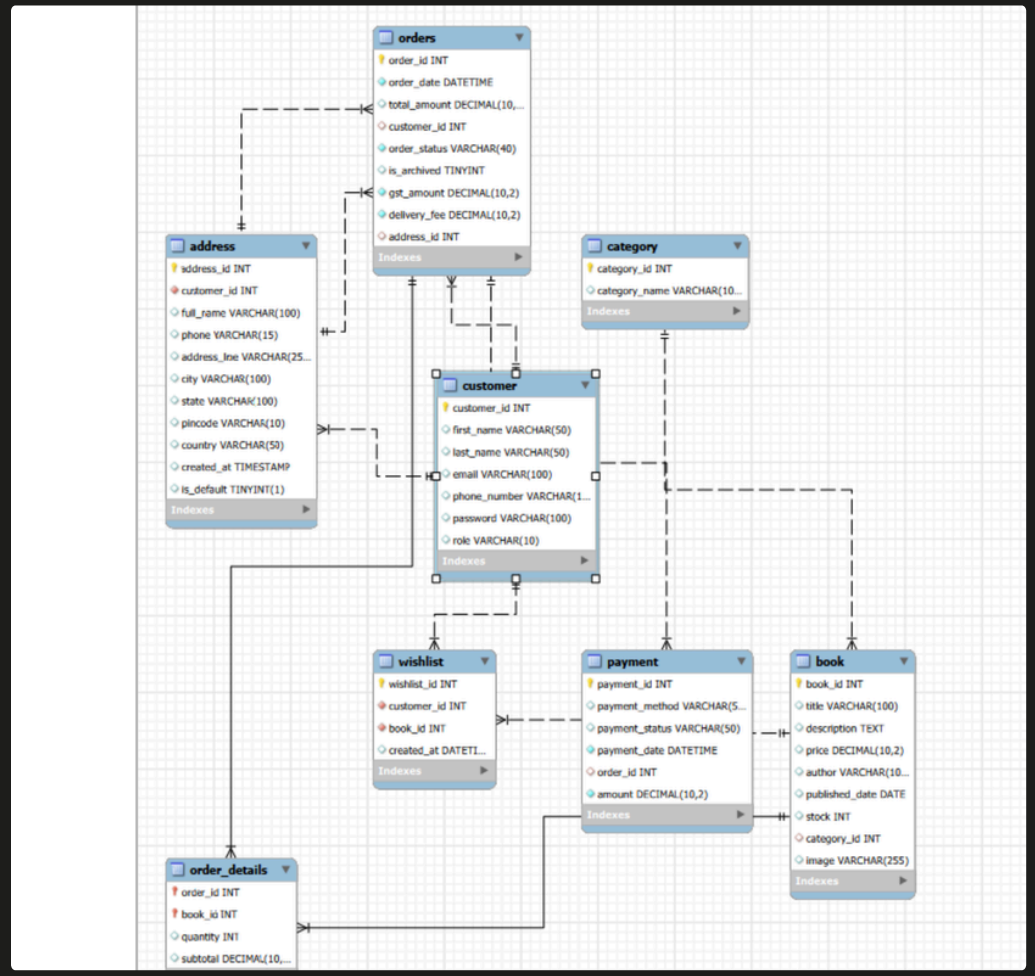
Real-Time Operations

Live cart, order placement, and payment processing with transaction support

Admin Control

Dedicated panel to manage books, categories, orders, and users

ER Model & Normalization



Core Entities

- **Customer** — user profiles and authentication
- **Books** — title, author, price, stock
- **Orders** — order header with status
- **Order_Items** — junction table resolving M:N
- **Payment** — transaction records
- **Wishlist** — saved items per user
- **Category** — book classification

Schema normalized to **3NF** — eliminating transitive dependencies, reducing redundancy, and ensuring consistency.

04 — IMPLEMENTATION

Technology Stack

MySQL

Relational database with DDL, DML, DQL and **normalized schema (3NF)**; indexing on frequently queried columns for performance

PHP Backend

Server-side logic handling business rules, SQL execution, and API responses

Frontend Integration

HTML/CSS/JS interface connected to PHP via prepared statements for secure data exchange

Transactions

ACID-compliant transactions (commit/rollback) ensuring **data consistency and atomicity during order processing**

05 — ALGORITHM

Order Processing Workflow

Add to Cart | Checkout | Create Order | Insert Items | Process Payment

Each step is wrapped in a database transaction. If any step fails — insufficient stock, payment error — the entire transaction rolls back, guaranteeing data consistency.

06 — QUERY DEMO

Multi-Table JOIN in Action

Sample Query

```
SELECT c.name, b.title, o.status
FROM Orders o
JOIN Customer c ON o.customer_id = c.id
JOIN Order_Items oi ON o.id = oi.order_id
JOIN Books b ON oi.book_id = b.id;
```

What It Does

- Joins **4 tables** to produce a complete order summary
- Retrieves customer name, book title, and order status in one result set
- Demonstrates referential integrity through foreign key joins
- Eliminates the need for multiple round-trip queries

07 — RESULTS & IMPACT

Outcomes & Societal Relevance

System Results

- Efficient, redundancy-free data handling across all modules
- Real-time cart, order, and payment operations functioning correctly
- Accurate order processing with full transactional integrity
- Scalable schema supporting growth in users and inventory

Societal Impact

- Democratizes access to books and learning resources
- Promotes digital literacy and e-commerce adoption
- Provides a model for small businesses to manage data professionally

Team Contributions & Future Scope

Team Roles

Akshat
Backend & Integration

Abhishek
UI / Frontend

Bhanu
Database Design

Ali
Admin Module

Amit
Authentication

Future Enhancements

- **Recommendation Engine** — ML-based book suggestions
- **Mobile App** — Native iOS & Android experience
- **Cloud Integration** — AWS/GCP for scalability
- **Analytics Dashboard** — Sales and user behavior insights

CONCLUSION

Thank You

The**KnowledgeHubSystem**demonstrateshow core DBMS concepts — normalization, transactions, indexing, and relationaldesign—cometogethertobuildareliable, scalable, real-world application.

DBMS Applied

3NF, FK constraints, JOINs, Transactions

System Built

Full-stack bookstore with admin panel

Guided By

Dr. Nishtha Phutela